

Integration

Week 4D

Overview

Introduction to Integration

Key Ideas - Introduction

- *Antiderivatives, Primitive Functions and the Indefinite Integral*
- *Integration of Power of x where n is a Natural Number*
- *Applications of the Primitive Function*
- *Rule 1: Constant Multiplier Rule*
- *Rule 2: Sum Rule*
- *Integration of Power of x where n is an Integer or Fraction*
- *Simplifying the Integrand*

Key Idea – Introduction

By now, you are familiar with differentiating common functions and have practised a range of techniques. We now consider the reverse process **integration**.

That is, given a function $f(x)$, we ask: *which function $F(x)$ has $f(x)$ as its derivative?* Such a function $F(x)$ is called an **antiderivative or primitive of $f(x)$** .

This leads us to the concepts of an antiderivative or primitive functions or integration. An important question is how to find the integral of a function. It can be found by reversing the process of differentiation that is by finding antiderivatives. **Integration is the reverse process of differentiation.**

This idea is formalised in the following result, which states that integration and differentiation are inverse processes.

derivative of $F(x) = f(x) \Leftrightarrow$ integral of $f(x) \quad dx = F(x) + c$

$$\frac{d}{dx} F(x) = f(x) \Leftrightarrow \int f(x) dx = F(x) + c$$

A key consequence is to integrate a function, we use antiderivative to find its primitive function. However, only some functions have antiderivatives that can be expressed in a simple, exact form

Key Idea – Primitive and the Indefinite Integral

Let's look an example:

Suppose we have differentiated some function and obtained,

$$f(x) = x^2 + 7.$$

“Now we ask: What function $F(x)$ did I differentiate to get $f(x) = x^2 + 7$?”

We know that the derivative of x^3 is $3x^2$.

Now we need to get a coefficient of 1!

We know the derivative of $\frac{1}{3}x^3$ is x^2 .

So, we have an answer, the x^2 term came from $\frac{1}{3}x^3$.

We also know the derivative of $7x$ is 7.

Hence, $F(x)$ should be,

$$F(x) = \frac{1}{3}x^3 + 7x$$

However it may have been, $F(x) = \frac{1}{3}x^3 + 7x + 1$ or $F(x) = \frac{1}{3}x^3 + 7x - \sqrt{5}$.

Or, in fact, the ‘1’ or the ‘ $\sqrt{5}$ ’ could be any constant term.

Each of these functions is called a **primitive function** or an **indefinite integral** of $f(x) = x^2 + 7$.

In fact, any function of the form

$$F(x) = \frac{1}{3}x^3 + 7x + c$$

where c is some constant (number), satisfies the equation

$$\frac{d}{dx} \left(\frac{1}{3}x^3 + 7x + c \right) = x^2 + 7 \Leftrightarrow \int x^2 + 7 \, dx = \frac{1}{3}x^3 + 7x + c.$$

For a general function $f(x)$, the indefinite integral is

$$\int f(x) \, dx = F(x) + c.$$

- $f(x)$ is called **the integrand**
- dx indicates that we need to integrate with respect to the variable x . Other variables may be used as shown in the examples.
- $F(x)$ is called a **primitive function** or **indefinite integral** of the function $f(x)$.
- Because of the unknown constant, c , the primitive function $F(x)$ is known as the **indefinite integral** of the function $f(x)$.
- The indefinite integral asks the question, ‘**What function do I differentiate to get the integrand.**’

Note: You can always check any integration by differentiating the result, $F(x)$, to see if it returns to $f(x)$.

Key Idea – Integration of Power of x

In general, the integral (or primitive function or antiderivative) of $f(x) = x^n$, where n is a real number, is given by

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$$

Examples: *Each of the examples below is asking the same question.*

The notation to be used for the primitive function will depend on the notation given for the derivative function.

Find the primitive function of $\frac{dy}{dx} = x^5$.

$$y = \int x^5 dx = \frac{x^6}{6} + c$$

Find $\int x^5 dx$.

$$\int x^5 dx = \frac{x^6}{6} + c$$

Find the antiderivative of $f(x) = x^5$.

$$F(x) = \int f(x) dx = \int x^5 dx = \frac{x^6}{6} + c$$

Find the indefinite integral of x^5 .

$$\int x^5 dx = \frac{x^6}{6} + c$$

Exercise:

(a) Find the primitive function of $\frac{dy}{dx} = x^2$

$$\frac{x^3}{3} + C$$

(b) Find the antiderivative of $f(x) = x^{12}$

$$\frac{x^{13}}{13} + C$$

(c) Find $\int x^{1.2} dx$

$$= \frac{x^{1.2+1}}{2.2} + C$$

$$= \frac{x^{2.2}}{2.2} + C$$

Examples: Find the primitive function of $\frac{dy}{dx} = 3$.

If we want to use, $\frac{x^{n+1}}{n+1} + c$ then we need to rewrite 3 as $3x^0$.

$$\begin{aligned} y = \int 3 \, dx &= \int 3x^0 \, dx \\ &= \frac{3x^{0+1}}{0+1} + c \\ &= 3x + c \end{aligned}$$

In general, the integral (or primitive function or antiderivative) of $f(x) = k$, where k is a real number, is given by

$$\int k \, dx = kx + c$$

Key Idea – Application of the Primitive Function

The primitive function is used to find the original function from its derivative

Example: The gradient function to a curve is given by $\frac{dy}{dx} = 3x^2 - x + 1$ and the curve passes through the point (2, 13), find the equation of the curve.

Step 1: Find primitive function:

$$\frac{dy}{dx} = 3x^2 - x + 1$$
$$\therefore y = \int 3x^2 - x + 1 \, dx = x^3 - \frac{x^2}{2} + x + c$$

Step 2: Use the given condition to evaluate constant:

Since the curve passes through (2, 13) $\therefore f(2) = 13$

$$(2)^3 - \frac{2^2}{2} + 2 + c = 13$$

$$8 - 2 + 2 + c = 13 \Rightarrow c = 13 - 8 \quad \therefore c = 5$$

Step 3: Write the final function:

$$y = x^3 - \frac{x^2}{2} + x + 5$$

Exercise:

Given $\frac{dy}{dx} = 3 + 2x - x^2$ and $f(2) = 5$, find the value of $f(x)$.

$$f(x) = y = \int 3 + 2x - x^2$$

$$y = 3x + \frac{2x^2}{2} - \frac{x^3}{3} + C$$

$$5 = 3 \times 2 + 2^2 - \frac{2^3}{3} + C$$

$$5 = 6 + 4 - \frac{8}{3} + C$$

$$5 = 10 - \frac{8}{3} + C$$

$$5 = \frac{22}{3} + C$$

$$C = 5 - \frac{22}{3}$$

$$C = -\frac{7}{3}$$

$$\begin{array}{ccc} f(2) = 5 & \uparrow & \uparrow \\ & x=2 & y=5 \end{array}$$

$$\therefore y = 3x + x^2 - \frac{x^3}{3} - \frac{7}{3}$$

Key Idea – Rule 1 and Rule 2

The following two rules mirror the rules for differentiation.

Rule 1 - Constant multiplier rule

If $k \in R$ is a constant, then

$$\int kf(x) \, dx = k \int f(x) \, dx$$

Rule 2 - Sum rule

$$\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx + \int g(x) \, dx$$

Note: $\int f(x) \times g(x) \, dx \neq \int f(x) \, dx \times \int g(x) \, dx$

Examples: Using Rule 1 and 2.

Rule 1:

$$\int 3x^4 dx = 3 \int x^4 dx \\ = \frac{3x^5}{5} + c$$

$$\int \frac{2x^3}{5} dx = \int \frac{2}{5} x^3 dx \\ = \frac{2}{5} \int x^3 dx \\ = \frac{2}{5} \times \frac{x^4}{4} + c \\ = \frac{x^4}{5} + c$$

Rule 2:

$$\int (3x^5 + 4x - 1) dx = \int 3x^5 dx + \int 4x dx + \int 1 dx \\ = \frac{3x^6}{6} + \frac{4x^2}{2} - x + c \\ = \frac{x^6}{2} + 2x^2 - x + c$$

Exercise:

Find the following:

$$(a) \int \left(2x^5 + \frac{x}{2} \right) dx = \int 2x^5 + \frac{x}{2} dx$$

$$= \frac{2x^6}{6} + \frac{x^2}{2 \times 2} + c$$

$$= \frac{x^6}{3} + \frac{x^2}{4} + c$$

$$(b) \int \left(x^2 + 3x - \frac{2}{3} \right) dx = \frac{x^3}{3} + \frac{3x^2}{2} - \frac{2}{3}x + c$$

Key Idea – Integration of Power of x where n is an Integer or Fraction

Integration of Powers of x where n is an Integer

Note: Rewrite reciprocal indices as negative indices

If the variable is in the denominator, express it with a negative index then differentiate.

$$\frac{1}{a^n} = a^{-n}$$

For example, $\frac{1}{a^4} = a^{-4}$

Examples: Find the indefinite integral of $\frac{1}{x^3}$, writing your answer with positive indices.

$$\begin{aligned}\int \frac{1}{x^3} dx &= \int x^{-3} dx \\ &= \frac{x^{-3+1}}{-3+1} + c \\ &= \frac{x^{-2}}{-2} + c = -\frac{1}{2x^2} + c\end{aligned}$$

Examples: Find $\int \frac{2}{3x^4} dx$, writing your answer with positive indices.

First, we need to rewrite the reciprocal index as negative index.

$$\begin{aligned}\int \frac{2}{3x^4} dx &= \frac{2}{3} \int x^{-4} dx \\ &= \frac{2}{3} \times \frac{x^{-4+1}}{-4+1} + c \\ &= \frac{2}{3} \times \frac{x^{-3}}{-3} + c \\ &= -\frac{2}{9x^3} + c\end{aligned}$$

Exercise:

Find the following, writing your answer with positive indices:

$$\begin{aligned}\text{(a)} \quad \int x^3 + x^{-2} \, dx &= \frac{x^{3+1}}{3+1} + \frac{x^{-2+1}}{-2+1} + C \\ &= \frac{x^4}{4} + \frac{x^{-1}}{-1} + C \\ &= \frac{x^4}{4} - \frac{1}{x} + C\end{aligned}$$

$$\begin{aligned}\text{(b)} \quad \int \frac{5}{x^3} - 4x + 1 \, dx &= \int 5x^{-3} - 4x + 1 \\ &= \frac{5x^{-3+1}}{-3+1} - \frac{4x^{1+1}}{1+1} + x + C \\ &= \frac{5x^{-2}}{-2} - \frac{4x^2}{2} + x + C \\ &= -\frac{5}{2x^2} - 2x^2 + x + C\end{aligned}$$

Integration of Powers of x where n is a Fraction

Like in differentiation, when the power of x is a fraction it is necessary to rearrange the function before integrating; in particular, there are three important cases to remember:

1. Rewrite surds (roots) as indices.

Expressions involving surds can be written with fractional indices using:

$$a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a} \right)^m$$

2. Rewrite reciprocal surds as negative indices.

If the variable is in the denominator, express it with a negative index

$$\frac{1}{a^{m/n}} = a^{-m/n},$$

3. Apply index laws before differentiating (multiplication/division).

When expressions involve multiplication or division of powers, simplify first using index laws:

$$a^n \times a^m = a^{n+m} \quad a^n \div a^m = a^{n-m}$$

Note: You need to understand the difference between leaving your answer in positive index form and surd form and you must be able to write both forms. In the exam, the question will tell you which form to use for your final answer.

For example,

- $x^{-\frac{1}{3}}$ is in negative index form
- $\frac{1}{x^{\frac{1}{3}}}$ is in positive index form
- $\frac{1}{\sqrt[3]{x}}$ is in surd form
-

Remember:

For example, $\frac{1}{3}x^{\frac{2}{3}} = \frac{x^{\frac{2}{3}}}{3}$ and $\frac{3}{5}x^2 = \frac{3x^2}{5}$

Examples: Find the following, writing your answer in surd form:

(a) $\int x^{\frac{4}{5}} dx$

$$\int x^{\frac{4}{5}} dx = \frac{x^{\frac{4}{5}+1}}{\frac{4}{5}+1} + c$$

$$= \frac{x^{\frac{9}{5}}}{\frac{9}{5}} + c$$

$$= \frac{5x^{\frac{9}{5}}}{9} + c$$

$$= \frac{5\sqrt[5]{x^9}}{9} + c \text{ (writing the answer with a surd)}$$

$$(b) \quad \int -\sqrt[3]{x^2} \, dx$$

Firstly, we need to rewrite the expression in index form.

$$\int -\sqrt[3]{x^2} \, dx = \int -x^{-\frac{2}{3}} \, dx$$

$$= \frac{x^{\frac{1}{3}}}{\frac{1}{3}} + c$$

$$= 3x^{\frac{1}{3}} + c$$

$$= \sqrt[3]{x} + c \text{ (writing the answer with a surd)}$$

(c) $\int \frac{2}{\sqrt[5]{x^6}} dx$

Firstly, we need to rewrite the expression in index form.

$$\begin{aligned}\int \frac{2}{\sqrt[5]{x^6}} dx &= 2 \int x^{-\frac{6}{5}} dx \\ &= \frac{2x^{-\frac{6}{5}+1}}{-\frac{6}{5}+1} + c = \frac{2x^{-\frac{1}{5}}}{-\frac{1}{5}} + c \\ &= -10x^{-\frac{1}{5}} + c \\ &= -\frac{10}{x^{\frac{1}{5}}} + c \\ &= -\frac{10}{\sqrt[5]{x}} + c\end{aligned}$$

$$(d) \int \frac{x}{\sqrt[3]{x}} dx$$

Firstly, simplify using index laws:

$$\begin{aligned} \int \frac{x}{\sqrt[3]{x}} dx &= \int x^{\frac{2}{3}} dx \\ &= \frac{x^{\frac{5}{3}}}{\frac{5}{3}} + c \\ &= \frac{3x^{\frac{5}{3}}}{5} + c \\ &= \frac{3\sqrt[3]{x^5}}{5} + c \end{aligned}$$

Exercises:

Integrate the following, write your answer in surd form:

$$\begin{aligned} \text{(a)} \quad \int \frac{1}{2\sqrt{x}} + \frac{5}{x^2} dx &= \int \frac{1}{2} x^{-\frac{1}{2}} + 5x^{-2} dx \\ &= \frac{1}{2} \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + \frac{5x^{-1}}{-1} + C \\ &= x^{\frac{1}{2}} - 5x^{-1} + C \\ &= \sqrt{x} - \frac{5}{x} + C \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \int 3x - \frac{2}{\sqrt[3]{x^5}} dx &= \int 3x - 2x^{-\frac{5}{3}} dx \\ &= \frac{3x^2}{2} - \frac{2x^{-\frac{2}{3}}}{-\frac{2}{3}} + C \\ &= \frac{3x^2}{2} + \frac{3}{\sqrt[3]{x^2}} + C \end{aligned}$$

Exercises:

Find the indefinite integral of the following, writing your answer in positive index form:

$$\begin{aligned} \text{(a)} \quad \int 2x^2 \sqrt{x} \, dx &= \int 2x^2 \times x^{\frac{1}{2}} \, dx \\ &= \int 2x^{2+\frac{1}{2}} \, dx \\ &= \int 2x^{\frac{5}{2}} \, dx \\ &= \frac{2x^{\frac{7}{2}}}{\frac{7}{2}} + C \\ &= \frac{4}{7} x^{\frac{7}{2}} + C \\ &= \frac{4}{7} \sqrt{x^7} + C \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \int \sqrt{x} \div x^{1/3} \, dx &= \int x^{\frac{1}{2}} \div x^{\frac{1}{3}} \, dx \\ &= \int x^{\frac{1}{2}-\frac{1}{3}} \, dx \\ &= \int x^{\frac{1}{6}} \, dx \\ &= \frac{6x^{\frac{7}{6}}}{\frac{7}{6}} + C \\ &= \frac{6}{7} \sqrt[6]{x^7} + C \end{aligned}$$

Key Idea – Simplifying Integrands

Sometimes an integrand can be simplified before the integral is evaluated. Simplification might be by using algebraic manipulations, or trigonometric identities.

Example: Determine $\int \frac{x^3 + 5}{3x^2} dx$.

First separate the algebraic fraction into two simpler fractions

$$\begin{aligned}\int \frac{x^3 + 5}{3x^2} dx &= \int \frac{x^3}{3x^2} dx + \int \frac{5}{3x^2} dx \\ &= \int \frac{x}{3} dx + \int \frac{5}{3x^2} dx \\ &= \int \frac{x}{3} dx + \int \frac{5}{3} x^{-2} dx \\ &= \frac{x^2}{3 \times 2} + \frac{5x^{-1}}{3 \times (-1)} + c \\ &= \frac{x^2}{6} - \frac{5}{3x} + c \quad \text{or} \quad \frac{1}{6}x^2 - \frac{5}{3x} + c\end{aligned}$$

Exercise:

Integrate the following:

(a) $\int (x-1)(2x+3)^2 dx$

$$\int (x-1)(4x^2+12x+9) dx$$

$$\int 4x^3+12x^2+9x-4x^2-12x+9 dx$$

$$\int 4x^3+8x^2-3x+9 dx$$

$$\frac{4x^4}{4} + \frac{8x^3}{3} - \frac{3x^2}{2} + 9x + C$$

$$x^4 + \frac{8}{3}x^3 - \frac{3}{2}x^2 + 9x + C$$

(b) $\int \frac{x^6-9x^5+6x^4}{x^3} dx$

$$= \int \frac{x^6}{x^3} - \frac{9x^5}{x^3} + \frac{6x^4}{x^3} dx$$

$$= \int x^3 - 9x^2 + 6x dx$$

$$= \frac{x^4}{4} - \frac{9x^3}{3} + \frac{6x^2}{2} + C$$

$$= \frac{x^4}{4} - 3x^3 + 3x^2 + C$$